

Financial Ratios — Detailed Reference Guide

For use with Empirical Project (ProjectFinRatio2026) and Final Exam Preparation
Financial statements required: 3 consecutive years (2022–2024)

Overview: The Five Categories of Financial Ratios

#	Category	Primary Question	Key Statements Used
1	Liquidity	Can the firm meet short-term obligations?	Balance Sheet
2	Solvency (Leverage)	Can the firm meet long-term obligations?	Balance Sheet + Income Statement
3	Profitability	How efficiently does the firm generate profit?	Income Statement + Balance Sheet
4	Efficiency (Activity)	How well does the firm use its assets?	Income Statement + Balance Sheet
5	Market Valuation	What does the market think the firm is worth?	Market Data + Financial Statements

Category 1: Liquidity Ratios

Measure the firm's ability to cover **short-term liabilities** with **short-term assets**.
Higher is generally better — but excessively high ratios may signal idle assets.

1.1 Current Ratio

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Benchmark	Interpretation
> 2.0	Strong liquidity; ample buffer
1.0 – 2.0	Acceptable; industry-dependent
< 1.0	Potential short-term solvency risk

- **Current Assets:** Cash, marketable securities, accounts receivable, inventory, prepaid expenses
- **Current Liabilities:** Accounts payable, accrued expenses, short-term debt, current portion of long-term debt
- **Limitation:** Includes inventory, which may not be quickly convertible to cash

1.2 Quick Ratio (Acid-Test Ratio)

$$\text{Quick Ratio} = \frac{\text{Cash} + \text{Marketable Securities} + \text{Accounts Receivable}}{\text{Current Liabilities}}$$

Equivalently:

$$\text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventory} - \text{Prepaid Expenses}}{\text{Current Liabilities}}$$

Benchmark	Interpretation
1.0	Can meet obligations without selling inventory
< 1.0	Dependent on inventory liquidation to cover short-term debt

- **Why it matters:** A more conservative measure than the current ratio; strips out illiquid assets

1.3 Cash Ratio

$$\text{Cash Ratio} = \frac{\text{Cash} + \text{Cash Equivalents}}{\text{Current Liabilities}}$$

- The most conservative liquidity measure
- Benchmark: varies widely by industry; generally 0.5–1.0 is considered adequate
- Useful for assessing worst-case liquidity (e.g., during a credit crunch)

1.4 Net Working Capital (NWC)

$$\text{NWC} = \text{Current Assets} - \text{Current Liabilities}$$

- Not a ratio, but a key liquidity metric
 - Positive NWC = operational buffer; Negative NWC = potential distress
 - Trend analysis (year-over-year) is more informative than a single value
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Category 2: Solvency (Leverage) Ratios

Measure long-term financial stability and the **degree of financial risk** from debt financing.

Higher leverage = higher risk but potentially higher return on equity (financial leverage effect).

2.1 Debt-to-Equity Ratio (D/E)

$$\text{D/E Ratio} = \frac{\text{Total Debt}}{\text{Total Shareholders' Equity}}$$

or more broadly:

$$\text{D/E Ratio} = \frac{\text{Total Liabilities}}{\text{Total Shareholders' Equity}}$$

Benchmark	Interpretation
< 1.0	Equity-financed; lower financial risk
1.0 – 2.0	Moderate leverage; typical for manufacturing
> 2.0	Highly leveraged; higher risk; common in banking/utilities

- **Policy implication:** A rising D/E trend signals increasing reliance on debt, potentially reducing financial flexibility
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2.2 Debt Ratio

$$\text{Debt Ratio} = \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

Benchmark	Interpretation
< 0.5 (50%)	More than half financed by equity — conservative

Benchmark	Interpretation
> 0.5 (50%)	More than half financed by debt — more leveraged

- Complements the D/E ratio; shows what proportion of assets are debt-financed

2.3 Equity Multiplier

$$\text{Equity Multiplier} = \frac{\text{Total Assets}}{\text{Total Shareholders' Equity}}$$

- Component of the **DuPont decomposition** (see Section 3.5)
- Higher value = greater leverage; firm using more debt relative to equity

2.4 Interest Coverage Ratio (Times Interest Earned)

$$\text{Interest Coverage} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

where EBIT = Earnings Before Interest and Taxes

Benchmark	Interpretation
> 3.0	Comfortable; can service debt well
1.5 – 3.0	Adequate but limited buffer
< 1.5	Potential difficulty meeting interest payments
< 1.0	EBIT does not cover interest — distress signal

- **Trend alert:** A declining interest coverage ratio over 3 years is a red flag for long-term solvency

2.5 Debt Service Coverage Ratio (DSCR)

$$\text{DSCR} = \frac{\text{Net Operating Income}}{\text{Total Debt Service}}$$

where Total Debt Service = Principal Repayment + Interest Expense

Benchmark	Interpretation
> 1.25	Typically required by lenders
= 1.0	Income exactly covers debt service — no margin

Benchmark	Interpretation
< 1.0	Negative cash flow after debt service — distress

Category 3: Profitability Ratios

Measure how efficiently the firm converts **revenue, assets, or equity into profit**.

3.1 Gross Profit Margin

$$\text{Gross Profit Margin} = \frac{\text{Revenue} - \text{COGS}}{\text{Revenue}} = \frac{\text{Gross Profit}}{\text{Revenue}}$$

- Reflects pricing power and production efficiency
 - Industry-specific: Software (~70–80%) vs. Grocery retail (~25–30%)
 - **Declining trend** may indicate rising input costs or pricing pressure
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3.2 Operating Profit Margin (EBIT Margin)

$$\text{Operating Margin} = \frac{\text{EBIT}}{\text{Revenue}}$$

- Captures profitability after operating expenses but **before** financing costs and taxes
 - Better comparability across firms with different capital structures
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3.3 Net Profit Margin

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Revenue}}$$

Benchmark	Interpretation
High margin	Strong pricing power, efficient cost control
Low margin	Thin profitability; vulnerable to revenue shocks

- Affected by tax policy, interest expense, non-recurring items
 - Compare with industry peers, not absolute benchmarks
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3.4 Return on Assets (ROA)

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total Assets}}$$

or using the DuPont link:

$$\text{ROA} = \text{Net Profit Margin} \times \text{Asset Turnover}$$

Benchmark	Interpretation
> 5%	Generally good; highly asset-heavy industries may score lower
< 2%	Poor asset utilization or thin margins

- Measures how well management deploys assets to generate profit

3.5 Return on Equity (ROE) — DuPont Decomposition

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

DuPont 3-Factor Decomposition:

$$\text{ROE} = \underbrace{\frac{\text{Net Income}}{\text{Revenue}}}_{\text{Net Profit Margin}} \times \underbrace{\frac{\text{Revenue}}{\text{Total Assets}}}_{\text{Asset Turnover}} \times \underbrace{\frac{\text{Total Assets}}{\text{Equity}}}_{\text{Equity Multiplier}}$$

Component	What It Captures
Net Profit Margin	Operational efficiency + pricing power
Asset Turnover	Efficiency of asset use
Equity Multiplier	Financial leverage

- **DuPont is essential for diagnosis:** two firms with the same ROE may arrive there via very different paths (one through high leverage, one through high margins)
- A rising ROE driven purely by rising Equity Multiplier = **riskier**, not better

3.6 Return on Invested Capital (ROIC)

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested Capital}}$$

where: - NOPAT = Net Operating Profit After Tax = $\text{EBIT} \times (1 - \text{Tax Rate})$ -
Invested Capital = Total Equity + Total Debt (or Total Assets – Non-interest-bearing Current Liabilities)

- Superior to ROE when comparing firms with different capital structures
 - $\text{ROIC} > \text{WACC} \rightarrow$ firm is creating value; $\text{ROIC} < \text{WACC} \rightarrow$ firm is destroying value
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3.7 EBITDA Margin

$$\text{EBITDA Margin} = \frac{\text{EBITDA}}{\text{Revenue}}$$

where EBITDA = Earnings Before Interest, Taxes, Depreciation & Amortization

- Proxy for **operating cash flow generation**
 - Used widely in valuation (EV/EBITDA multiple)
 - Strips out capital structure and accounting differences
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Category 4: Efficiency (Activity) Ratios

Measure how effectively the firm manages its **assets and operational cycle**.

4.1 Asset Turnover

$$\text{Asset Turnover} = \frac{\text{Revenue}}{\text{Average Total Assets}}$$

where Average Total Assets = $(\text{Beginning} + \text{Ending Total Assets}) \div 2$

Interpretation

Higher = better asset utilization per unit of revenue
Capital-intensive industries (steel, airlines) naturally have lower ratios
Retail/service firms typically have higher ratios

4.2 Inventory Turnover

$$\text{Inventory Turnover} = \frac{\text{COGS}}{\text{Average Inventory}}$$

$$\text{Days Inventory Outstanding (DIO)} = \frac{365}{\text{Inventory Turnover}}$$

Benchmark	Interpretation
High turnover / Low DIO	Efficient inventory management; less capital tied up
Low turnover / High DIO	Slow-moving inventory; risk of obsolescence

4.3 Accounts Receivable Turnover & Days Sales Outstanding (DSO)

$$\text{AR Turnover} = \frac{\text{Revenue}}{\text{Average Accounts Receivable}}$$

$$\text{DSO} = \frac{365}{\text{AR Turnover}} = \frac{\text{Average AR}}{\text{Revenue}/365}$$

- Lower DSO = faster collections = better cash flow
- Rising DSO over 3 years → collections are weakening or credit terms are loosening

4.4 Accounts Payable Turnover & Days Payable Outstanding (DPO)

$$\text{AP Turnover} = \frac{\text{COGS}}{\text{Average Accounts Payable}}$$

$$\text{DPO} = \frac{365}{\text{AP Turnover}}$$

- Higher DPO = company is taking longer to pay suppliers = using supplier credit as free financing
 - Too high DPO may signal cash problems or strained supplier relationships
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4.5 Cash Conversion Cycle (CCC)

$$\text{CCC} = \text{DIO} + \text{DSO} - \text{DPO}$$

CCC Value	Interpretation
Low / Negative	Very efficient; company collects before it pays (e.g., Amazon)
High	Capital tied up in operations for a long time

- The CCC represents the number of days a firm needs to finance its own operating cycle
- **Ideal target:** Minimize CCC (reduce DIO and DSO; extend DPO)

Category 5: Market Valuation Ratios

Link financial performance to **market price**, used for investment decisions.

5.1 Price-to-Earnings Ratio (P/E)

$$\text{P/E} = \frac{\text{Market Price per Share}}{\text{Earnings per Share (EPS)}}$$

$$\text{EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Weighted Average Shares Outstanding}}$$

P/E	Interpretation
High P/E	Market expects high future growth; or stock may be overvalued
Low P/E	Market expects slow growth; or stock may be undervalued
Negative P/E	Company is unprofitable

- Use **trailing P/E** (last 12 months EPS) or **forward P/E** (estimated next 12 months)
- Compare to sector peers and historical average, not absolute levels

5.2 Price-to-Book Ratio (P/B)

$$P/B = \frac{\text{Market Price per Share}}{\text{Book Value per Share}}$$

$$\text{Book Value per Share} = \frac{\text{Total Equity}}{\text{Shares Outstanding}}$$

P/B	Interpretation
< 1.0	Market values firm below book; may signal distress or deep value
1.0 – 3.0	Typical range for many industries
> 3.0	Strong intangible assets or growth premium

- Less meaningful for firms with significant intangible assets (tech, pharma)

5.3 Enterprise Value Multiples

$$EV = \text{Market Capitalization} + \text{Total Debt} - \text{Cash \& Equivalents}$$

$$EV/EBITDA = \frac{EV}{EBITDA}$$

$$EV/\text{Revenue} = \frac{EV}{\text{Revenue}}$$

- EV/EBITDA is the most common valuation multiple in M&A and equity research
- Capital-structure neutral: useful for comparing firms with different debt levels
- General benchmark: EV/EBITDA of 8–12x is common for mature firms; higher for growth firms

5.4 Dividend Yield & Payout Ratio

$$\text{Dividend Yield} = \frac{\text{Annual Dividend per Share}}{\text{Market Price per Share}}$$

$$\text{Payout Ratio} = \frac{\text{Dividends Paid}}{\text{Net Income}} = \frac{\text{DPS}}{\text{EPS}}$$

Payout Ratio	Interpretation
0–30%	Growth-oriented; retaining most earnings
30–60%	Balanced; common for mature companies
> 70%	High income distribution; limited reinvestment; may be unsustainable
> 100%	Paying dividends out of reserves — red flag

Synthesis: The DuPont Framework (Master Diagnostic Tool)

ROE		
Net Profit Margin (Profitability)	Asset Turnover (Efficiency)	Equity Multiplier (Leverage)
Net Income / Revenue	Revenue / Avg. Assets	Total Assets / Equity

How to use it in your empirical project: 1. Calculate ROE for each year (2022, 2023, 2024) 2. Decompose into all three components 3. Identify which driver is responsible for ROE changes 4. Connect to company strategy or industry conditions

How to Structure Your Empirical Project Analysis

Step 1: Company Profile

- Name, industry, stock exchange listing
- Why the company is interesting (competitive position, recent events, sector dynamics)
- Brief description of business model

Step 2: Financial Statements (Attach)

- Income Statement (2022, 2023, 2024)
- Balance Sheet (2022, 2023, 2024)
- Cash Flow Statement (optional but recommended)

Step 3: Calculate and Present Ratios

Organize in a table:

Ratio	Formula	2022	2023	2024	Trend
Current Ratio	CA / CL				↑ / ↓ / →
Quick Ratio	(CA-Inv) / CL				
D/E Ratio	Total Debt / Equity				
Interest Coverage	EBIT / Interest				
Gross Margin	Gross Profit / Revenue				
Net Margin	Net Income / Revenue				
ROA	Net Income / Assets				
ROE	Net Income / Equity				
Asset Turnover	Revenue / Avg. Assets				
Inventory Turnover	COGS / Avg. Inventory				
DSO	AR / (Revenue/365)				
CCC	DIO + DSO - DPO				
P/E Ratio	Price / EPS				
P/B Ratio	Price / Book				

Step 4: Interpret Results

For each ratio category, answer: - **What is the current level?** (high/low relative to industry) - **What is the trend?** (improving, deteriorating, stable) - **What does this imply?** (operational, strategic, or risk implication)

Step 5: Conclusion & Policy Implications

Synthesize across all five categories:

Category	Overall Assessment	Implication
Liquidity	Strong / Adequate / Weak	Risk of short-term default?
Solvency	Conservative / Moderate / Leveraged	Long-term sustainability?
Profitability	Growing / Stable / Declining	Competitive advantage?
Efficiency	Improving / Stable / Declining	Operational issues?
Valuation	Overvalued / Fair / Undervalued	Investment signal?

Policy implication examples: - If leverage is rising and interest coverage falling → firm should consider equity issuance or asset disposal - If DSO is

rising → review credit policy; consider stricter collection procedures - If ROE is falling despite stable margins → leverage is declining (deleveraging); or asset base growing faster than income

Quick Reference: Key Formulas at a Glance

Liquidity

Ratio	Formula
Current Ratio	$CA \div CL$
Quick Ratio	$(CA - \text{Inventory} - \text{Prepaid}) \div CL$
Cash Ratio	$\text{Cash} \div CL$
Net Working Capital	$CA - CL$

Solvency

Ratio	Formula
Debt-to-Equity	$\text{Total Debt} \div \text{Equity}$
Debt Ratio	$\text{Total Liabilities} \div \text{Total Assets}$
Equity Multiplier	$\text{Total Assets} \div \text{Equity}$
Interest Coverage	$\text{EBIT} \div \text{Interest Expense}$
DSCR	$\text{NOI} \div (\text{Principal} + \text{Interest})$

Profitability

Ratio	Formula
Gross Margin	$\text{Gross Profit} \div \text{Revenue}$
Operating Margin	$\text{EBIT} \div \text{Revenue}$
Net Margin	$\text{Net Income} \div \text{Revenue}$
ROA	$\text{Net Income} \div \text{Avg. Total Assets}$
ROE	$\text{Net Income} \div \text{Equity}$
ROIC	$\text{NOPAT} \div \text{Invested Capital}$
DuPont ROE	$\text{Net Margin} \times \text{Asset Turnover} \times \text{Equity Multiplier}$

Efficiency

Ratio	Formula
Asset Turnover	$\text{Revenue} \div \text{Avg. Total Assets}$
Inventory Turnover	$\text{COGS} \div \text{Avg. Inventory}$

Ratio	Formula
DIO	$365 \div \text{Inventory Turnover}$
AR Turnover	$\text{Revenue} \div \text{Avg. AR}$
DSO	$365 \div \text{AR Turnover}$
AP Turnover	$\text{COGS} \div \text{Avg. AP}$
DPO	$365 \div \text{AP Turnover}$
CCC	DIO + DSO – DPO

Market Valuation

Ratio	Formula
EPS	$(\text{Net Income} - \text{Preferred Div.}) \div \text{Shares Outstanding}$
P/E	$\text{Market Price} \div \text{EPS}$
P/B	$\text{Market Price} \div \text{Book Value per Share}$
EV	$\text{Mkt Cap} + \text{Total Debt} - \text{Cash}$
EV/EBITDA	$\text{EV} \div \text{EBITDA}$
Dividend Yield	$\text{DPS} \div \text{Market Price}$
Payout Ratio	$\text{Dividends} \div \text{Net Income}$

Common Red Flags in Ratio Analysis

Signal	Possible Interpretation
Current Ratio < 1.0 AND declining	Liquidity deterioration; may default on short-term obligations
D/E rising + Interest Coverage falling	Debt burden increasing faster than earnings
Net Margin declining despite revenue growth	Cost inflation or pricing pressure
ROE rising but driven solely by leverage (Equity Multiplier)	Risk is increasing, not real profitability improvement
DSO rising year-over-year	Collection problems; accounts receivable quality declining
DIO rising year-over-year	Slowing demand or inventory management issues
Payout Ratio > 100%	Dividends are unsustainable; being funded from reserves or debt
ROIC < WACC	Firm is destroying shareholder value despite being “profitable”